Assignment 5

Q. Polymorphism means 'many forms'. In OOP, the same method name (message) is sent to different object (of related classes), but each object responds to it in own way – implemented differently.

Eg. i) withdraw() on Bank Account

Same Message : Withdraw() is sent all the subclasses of Bank Account

+ Saving Acc +	+ Current Acc
withdraw(){ checks minimum balance before deducting };	withdraw() { checks Overdraft limit before deducting };

Result : But Implemented Differently

Same message 'withdraw()' sent to BankAccount, but each subclass implements it differently.

ii) makeSound() on Animal

+ Dog +	+ Cat +
makeSound() { print 'Bark Bark!' };	makeSound() { print 'Meow, Meow!' };

+ Cow +

makeSound() { print Moo, Moo! }

iii) getArea() on Shape

* Circle *	* Rectangle *
getArea() { return 3.14 * r * r }	getArea() { return length * width }

* Triangle *

getArea() { return 0.5 * base * height }

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iv) CalculateSalary() on Employee

+ Manager +
CalculateSalary() { base Salary + teambonus + allowance } ;

+ Developer +
CalculateSalary() { base Salary + project bonus } ;

* Intern *
Calculate Salary() { fixed Stipend only, ono bonus } ;

v) Pay() on Payment

+ UPI Payment +
Pay() { validate UPI PIN and deduct from linked Account } ;

+ Credit Card +
Pay() { authorize card, deduct from credit line } ;

+ Cash Payment
pay() { physically count and hand over cash } ;

vi) login() on User

+ Admin +
login() { Verify credentials + 2FA OTP required } ;

+ Customer +
login() { Verify username and password only } ;

+ Guest +
login() { auto-login with limited access, No credentials } ;

vii) move() on Vehicle

+ Car +
move() { Start Engine, engage, Press Accelerator

+ Bicycle +
move() { pedal the wheel manually } ;

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viii) generateReport() on Department

+ Finance Dept +
generateReport() { produce profile balance sheet };

+ HR Dept + +
generateReport() { produce employee attendance & payroll report };

* IT Dept + +
generateReport() { Produce Server uptime and bug tracking Report };

ix) SendNotification() on Notification Service

+ Email Service +
SendNotification() { Compose and Send via SMTP email server };

+ SMS Service +
SendNotification() { Send Text message via telecom gateway };

x) Product Complaint / File Complaint on Customer Support

+ Product Complaint +
fileComplain() { assign to product quality team + generate replacement order };

+ Delivery Complaint +
File Complaint() { assign to logisticks team + track shipment };

+ Billing Complaint +
file Complaint() { assign to finance team + review transaction };